

# Marketing Agency CMS - Backend Roadmap

Implementation Plan: Database Schemas & express-validator Logic

Repository: [github.com/mohamedahmedzaghlol/Marketing-Agency-CMS](https://github.com/mohamedahmedzaghlol/Marketing-Agency-CMS)

## Week 1: Category Model

### Data Fields:

- **name:** String, Required, Unique.
- **slug:** String, Unique (Generated from name).
- **description:** String, Optional.
- **colorCode:** String (Hex format for UI tags).

### Logic & Validation:

- Full CRUD (Create, Read, Update, Delete).
- **express-validator:** Check for non-empty name, valid hex code, and slug format.

## Week 2: User Model & Authentication

### Data Fields:

- **username:** String, Required, Unique.
- **email:** String, Required, Unique.
- **password:** String, Required (Bcrypt Hashed).
- **profileImage:** String (File path for Multer).
- **role:** Enum ['Admin', 'Editor'].

### Logic & Validation:

- User Registration and Login with JWT.
- **express-validator:** `isEmail()`, `isLength({min: 8})` for password, `normalizeEmail()`.

## Week 3: Post Model (Core Content)

### Data Fields:

- **title:** String, Required (50-60 chars for SEO).
- **body:** String (HTML/Rich Text).
- **excerpt:** String (Short summary).
- **featuredImage:** String (Required path).
- **status:** Enum ['Draft', 'Published', 'Scheduled'].
- **category:** ObjectId (Ref: Category).
- **author:** ObjectId (Ref: User).
- **scheduleDate:** Date (Used for scheduled posts).

### Logic & Validation:

- Advanced CRUD with image upload (Multer).
- **express-validator:** Check for valid ObjectIDs, validate title length, and date formats.

## Week 4: Contact & Newsletter Model

### Data Fields:

- **fullName:** String, Required.
- **email:** String, Required, Valid Format.
- **message:** String (Required for inquiries).

- **type:** Enum ['Inquiry', 'Newsletter'].

### Logic & Validation:

- Lead capturing and subscription logic.
- **express-validator:** Sanitize inputs, check for valid email, and body length.

*Note: All validation logic is implemented using **express-validator** middleware for consistent error handling and data sanitization.*